SKILL BOUNTY PLATFORM

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE AND ENGINEERING

Use Case Report

submitted by

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CERTIFICATE

This is to certify that the Use Case report entitled **"Skill Bounty Platform"** that is being submitted by **Karpurapu Meghana(23501A0564)**, as part of Assignment-2 for the **Blockchain Technology(23CS4701B)** course in **4-1** during the academic year **2026-27**.

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1. INTRODUCTION

1.1. Overview of Blockchain Technology

Blockchain technology is a decentralized and distributed digital ledger that enables transactions and records to be stored across a network of participating computers without requiring a single central authority. Unlike conventional centralized systems, where a trusted organization maintains and controls the database, blockchain distributes transaction records among multiple network participants. This decentralized architecture improves transparency, security, traceability, and resistance to unauthorized modification.

One of the most important developments in blockchain technology is the introduction of smart contracts. Smart contracts are self-executing programs deployed on a blockchain that automatically perform predefined operations when specified conditions are satisfied. They eliminate the need for intermediaries for many digital transactions and allow business rules to be enforced programmatically.

1.2. Relevance of Blockchain in Skill Bounty Platform

The growth of the digital economy has created numerous opportunities for individuals to provide technical, creative, and professional skills through online platforms. Freelancers and skilled individuals can work with clients from different geographical locations without being physically present in the same place.

However, conventional freelance and skill-marketplace platforms generally depend on centralized organizations to manage user profiles, job listings, applications, payments, work verification, dispute resolution, and reputation. This creates dependency on the platform and introduces challenges related to trust, transparency, payment security, and dispute management.

A blockchain-based skill bounty platform can address several of these limitations by recording important activities on a decentralized ledger. Smart contracts can be used to define the conditions of a work agreement, manage reward allocation, and release payments after the completion and approval of work.

The proposed SkillBounty platform applies blockchain technology to create a transparent and decentralized environment where clients can publish skill-based bounties and freelancers can discover and apply for suitable opportunities. The platform uses Solidity-based smart contracts to manage important job and payment operations.

2. BACKGROUND

Many freelancing and skill-based platforms rely on centralized systems to manage job listings, user profiles, payments, and work records. Freelancers may face difficulties in finding trustworthy opportunities, receiving timely payments, and resolving disputes, while clients may struggle to ensure reliable work completion. This creates trust, transparency, and payment-related challenges between clients and freelancers.

2.1 Existing Skill-Based Freelancing Systems

Traditional freelancing platforms provide an online environment where clients can post jobs and freelancers can offer their skills. These platforms generally manage user accounts, job listings, applications, payments, work submissions, reviews, and dispute resolution through centralized servers.

Although such platforms make it easier to connect clients and freelancers, users must depend on the platform for maintaining records, processing payments, managing reputation, and resolving disputes.

2.2 Limitations of Centralized Platforms

2.2.1 Centralized Control

A centralized organization controls the major operations of the platform, including user information, job records, payments, and reputation data. Users therefore depend on the platform to maintain accurate and accessible records.

2.2.2 Trust Issues

Clients need assurance that freelancers will complete the assigned work, while freelancers need assurance that clients will release the agreed payment after successful completion. This dependency creates trust-related challenges

2.2.3 Payment and Escrow Problems

Traditional payment systems may involve intermediaries, processing delays, transaction fees, and platform restrictions. Payment disputes may also require intervention from the platform.

2.2.4 Lack of Transparency

Users generally cannot independently verify how important platform operations are processed. Job status, payment processing, and other records are maintained within centralized databases.

2.2.5 Reputation Dependency

Freelancer ratings, reviews, and reputation are usually maintained by the centralized platform. This makes reputation information dependent on the platform's database and policies.

2.2.6 Dispute Resolution

Disputes may occur because of incomplete work, delayed delivery, differences in requirements, or disagreements regarding payment. Resolving such disputes generally requires intervention by the centralized platform.

2.3 Need for a Decentralized Approach

The limitations of centralized freelancing systems create a need for a platform that can provide greater transparency, secure payment handling, and verifiable transaction records.

Blockchain technology provides an alternative approach by distributing transaction records across a network and using cryptographic mechanisms to protect their integrity. Smart contracts can also automate predefined rules for job management and payment processing.

In a skill-based marketplace, blockchain can therefore be used to establish a more transparent relationship between clients and freelancers.

2.4 Blockchain-Based Skill Bounty Platform

The proposed SkillBounty platform applies blockchain technology to the skill marketplace model. Instead of depending entirely on a centralized authority, important operations are implemented through Solidity smart contracts deployed on a blockchain network.

The platform allows clients to create bounties, specify rewards and deadlines, and review applications from freelancers. Freelancers can browse available opportunities, submit bids, and complete selected jobs.

The major workflow is:

Create Bounty → Apply/Bid → Select Freelancer → Secure Reward → Complete Work → Submit Work → Approve Work → Release Reward

This approach improves transparency by recording important transactions on the blockchain while smart contracts enforce predefined conditions.

2.5 Role of Smart Contracts

Smart contracts form the core of the blockchain-based functionality of SkillBounty. They manage operations such as bounty creation, job management, bidding, work submission, payment handling, and user reputation.

The platform uses multiple Solidity contracts, including:

- **BountyBoard** – manages bounty creation and work submission.
- **JobMarket** – manages jobs, bids, freelancer selection, and job completion.
- **Escrow** – manages reward deposits, payment release, and refunds.
- **UserProfile** – manages profiles, skills, reviews, and reputation.
- **DisputeResolution** – supports dispute-related operations.
- **Governance** – provides governance-related functionality.
- **SubContracting** – manages subcontracting functionality.
- **VRTToken** – provides token-related functionality.

2.6 Decentralized Work Delivery

SkillBounty also integrates IPFS-based storage through Pinata for work deliverables. Freelancers can upload completed work through the platform, and the uploaded file is stored using decentralized storage.

The resulting IPFS reference can then be associated with the work submission, allowing the client to access and review the delivered work.

2.7 Proposed Solution

SkillBounty combines blockchain-based transaction management with a web-based skill marketplace. MetaMask is used for wallet interaction, Solidity smart contracts manage blockchain operations, Mantle Sepolia is used as the public test network, and IPFS through Pinata is used for decentralized work-file storage.

The proposed system aims to provide:

- Transparent job and payment records
- Secure blockchain-based transactions
- Smart-contract-based workflow automation
- Decentralized work delivery
- Skill and reputation management
- Reduced dependency on centralized intermediaries
- Improved trust between clients and freelancers

Thus, SkillBounty provides a practical blockchain-based approach for connecting skills with work opportunities while maintaining transparency and secure reward management.

3. BLOCKCHAIN BASICS

Blockchain technology is a distributed ledger system that ensures secure, transparent, and tamper-proof transactions. Unlike traditional systems that rely on intermediaries, blockchain enables trustless peer-to-peer interactions, reducing the need for third parties in financial transactions, supply chain management, and digital warranty verification. Its decentralized nature enhances security and transparency, making it an ideal solution for many industries. Below are the key concepts that define blockchain technology:

3.1. Decentralization

Blockchain is a decentralized technology where transaction records are maintained across multiple nodes instead of being controlled by a single authority. In SkillBounty, blockchain reduces dependency on a centralized platform by recording important job and payment transactions through smart contracts.

3.3 Immutability

Blockchain provides transparency because confirmed transactions can be verified on the network. In SkillBounty, activities such as job creation, bidding, freelancer selection, and reward transactions can be recorded on the blockchain, making the workflow more traceable.

3.3. Transparency

Once a blockchain transaction is confirmed, it cannot be easily modified or deleted. This helps SkillBounty maintain reliable records of important job and payment activities and reduces the possibility of unauthorized manipulation.

3.4. Smart Contracts

Smart contracts are programs deployed on a blockchain that automatically execute predefined rules when required conditions are satisfied. SkillBounty uses Solidity smart contracts to manage bounty creation, job management, bidding, escrow, work submission, payment release, and reputation-related operations.

3.5. Consensus Mechanisms

Blockchain networks rely on consensus mechanisms to validate transactions and maintain system integrity. The most widely used mechanism is Proof of Work (PoW), where miners solve complex mathematical puzzles to confirm transactions, as seen in Bitcoin. Another approach, Proof of Stake (PoS), selects validators based on the number of tokens they hold, which is used in Ethereum 2.0. A more efficient variation, Delegated Proof of Stake (DPoS), involves electing a smaller group of nodes to verify transactions, improving scalability and reducing energy consumption. These consensus models prevent malicious actors from manipulating the system and ensure that only valid transactions are recorded on the blockchain.

3.6. Cryptographic Security

Blockchain security relies on advanced cryptographic techniques to protect user data and transactions. Each transaction is authenticated using public and private keys, ensuring that only authorized individuals can initiate transactions. Hash functions further enhance security by converting data into fixed-length unique codes, making it nearly impossible to reverse-engineer original information. Additionally, blockchain uses encryption to protect sensitive data, ensuring that unauthorized users cannot access confidential records. This high level of security makes blockchain an ideal technology for digital warranties, financial transactions, and identity management.

By combining these key features—decentralization, immutability, transparency, smart contracts, and cryptographic security—blockchain provides a robust and efficient foundation for various industries, including digital warranty management. It ensures secure, automated, and verifiable transactions, reducing fraud and increasing trust in warranty claims and ownership transfers.

These fundamental blockchain principles make it secure, transparent, and efficient, enabling applications in finance, healthcare, supply chains, and digital warranties.

4. USE CASE OVERVIEW

The use case for a Blockchain-Based Skill Bounty Platform aims to transform traditional freelancing by leveraging blockchain technology. This system enables secure, transparent, and trust-based job creation, freelancer selection, work submission, and reward distribution through smart contracts.

4.1. Objectives

The main objective of SkillBounty is to develop a decentralized platform that connects clients with skilled individuals through blockchain-based bounties and secure reward management.

The major objectives are:

- To allow clients to create and publish skill-based bounties.
- To allow freelancers to browse and apply for suitable opportunities.
- To provide transparent freelancer selection and job management.
- To use smart contracts for secure and automated transactions.
- To provide escrow-based reward management.
- To allow freelancers to submit completed work.
- To use IPFS for decentralized work-file storage.
- To enable clients to approve completed work and release rewards.
- To maintain user skills, reviews, and reputation.
- To provide transparent and verifiable blockchain transaction records.

4.2. Scope of the System

The SkillBounty platform supports the complete lifecycle of a skill-based job, from creating a bounty to completing the work and releasing the reward.

The system includes two primary users:

Client: Creates bounties, specifies requirements and rewards, reviews applications, selects freelancers, and approves completed work.

Freelancer: Creates a profile, adds skills, browses available bounties, submits bids, completes assigned work, and receives rewards after approval.

The system uses MetaMask for wallet interaction, Solidity smart contracts for blockchain operations, Mantle Sepolia as the test network, and IPFS through Pinata for work deliverables.

4.3. System Architecture

The SkillBounty system consists of four major layers and consists of the following components:

4.3.1. User Layer

The user layer consists of clients and freelancers who interact with the platform using their blockchain wallets.

4.3.2. Web Application Layer

The application layer provides the web interface for creating bounties, browsing jobs, submitting bids, managing profiles, uploading work, and viewing job status.

4.3.3 Smart Contract Layer

The smart contract layer contains Solidity contracts responsible for job management, bidding, escrow, profiles, disputes, and related blockchain operations.

4.3.4 Blockchain and Storage Layer

The blockchain layer uses Mantle Sepolia for transaction processing and verification. IPFS through Pinata is used for storing work deliverables.

The overall architecture can be represented as:

Client/Freelancer → SkillBounty Web Application → MetaMask → Solidity Smart Contracts → Mantle Sepolia

Work File → SkillBounty → Pinata → IPFS

4.4 System Components

The major components of SkillBounty are:

4.4.1 User Profile

Manages user profiles, skills, reviews, endorsements, and reputation.

4.4.2 Bounty Management

Allows clients to create bounties with descriptions, rewards, deadlines, and requirements.

4.4.3 Job Market

Manages job listings, freelancer bids, freelancer selection, milestones, and job completion.

4.4.4 Escrow

Handles reward deposits, payment release, milestone payments, and refunds according to the smart-contract workflow.

4.4.5 Work Delivery

Allows freelancers to upload completed work and submit it for client review.

4.4.6 IPFS Storage

Stores submitted work through Pinata and provides an IPFS content identifier for accessing the uploaded file.

4.4.7 Dispute Resolution

Provides functionality for handling disagreements between clients and freelancers.

System Workflow

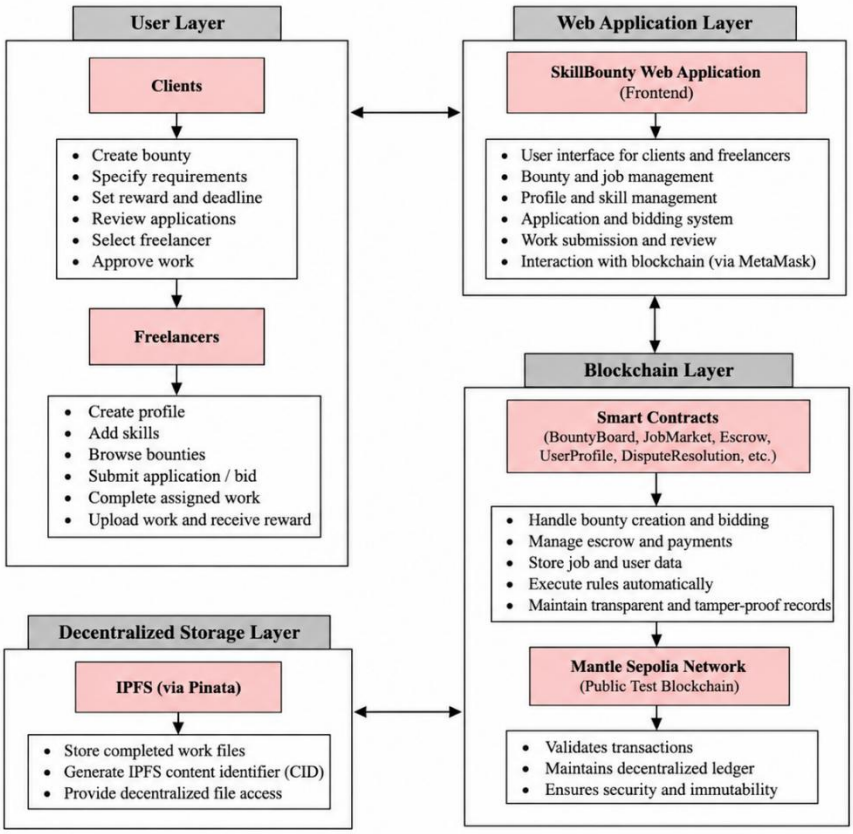


Fig 4.1: Architecture of Blockchain-Based Skill Bounty Platform (SkillBounty)

5.IMPLEMENTATION

5.1. Setting Up the Blockchain Environment

SkillBounty is implemented as a web-based decentralized application using blockchain technology. The frontend provides the user interface, while Solidity smart contracts manage the core blockchain operations.

The main technologies used are:

- **Frontend:** React.js, Vite, Tailwind CSS
- **Blockchain:** Mantle Sepolia Test Network
- **Smart Contracts:** Solidity
- **Development Framework:** Hardhat
- **Wallet:** MetaMask
- **Storage:** IPFS through Pinata
- **Blockchain Interaction:** Web3/Ethers-based integration

5.2. Defining Smart Contracts Implementation

The core functionality of SkillBounty is implemented using Solidity smart contracts. Different contracts are responsible for specific operations.

The major contracts include:

- **BountyBoard:** Creates bounties and manages work submissions.
- **JobMarket:** Handles job creation, bidding, freelancer selection, and completion.
- **Escrow:** Manages reward deposits, payment release, and refunds.
- **UserProfile:** Maintains user profiles, skills, reviews, and reputation.
- **DisputeResolution:** Supports dispute-related operations.
- **Governance:** Provides governance functionality.
- **SubContracting:** Supports subcontracting operations.
- **VRTToken:** Provides token-related functionality.

5.2.1. Core Smart Contract Implementation

The following selected code snippets show the core blockchain operations implemented in the JobMarket smart contract for job creation, bidding, freelancer selection, and escrow.

Code Snippet 5.1: Job Creation — createJob()

```
function createJob(
    string calldata title, string calldata description,
    string calldata category, uint256 budget, uint256 deadline
) external returns (uint256) {
    require(bytes(title).length > 0 && bytes(description).length > 0,
        "Title & desc required");
    require(budget > 0 && deadline > block.timestamp, "Bad budget/deadline");
    jobCounter++;
    uint256 jid = jobCounter;
    jobs[jid] = Job({
        id: jid, client: msg.sender, title: title,
        description: description, category: category, budget: budget,
        deadline: deadline, status: JobStatus.Open,
        selectedFreelancer: address(0), acceptedBidId: 0,
        createdAt: block.timestamp, deliveredAt: 0, ...
    });
    _ensureProfile(msg.sender);
    emit JobCreated(jid, msg.sender, title, budget, category);
    return jid;
}
```

Code Snippet 5.2: Placing a Bid — placeBid()

```
function placeBid(uint256 jobId, uint256 amount,
    uint256 completionDays, string calldata proposal)
    external nonReentrant returns (uint256) {
    Job storage job = jobs[jobId];
    require(job.id != 0, "Job !exist");
    require(job.status == JobStatus.Open, "Not open");
    require(job.client != msg.sender, "Own job");
    require(amount > 0, "Amount must be > 0");
    require(bytes(proposal).length > 0, "Proposal required");
    require(!hasBidOnJob[jobId][msg.sender], "Already bid");
    require(block.timestamp < job.deadline, "Deadline passed");
    bidCounter++;
    uint256 bid_ = bidCounter;
    bids[bid_] = Bid(bid_, jobId, msg.sender, amount,
        completionDays, proposal, block.timestamp, true);
    jobBids[jobId].push(bid_);
    hasBidOnJob[jobId][msg.sender] = true;
    _ensureProfile(msg.sender);
    emit BidPlaced(bid_, jobId, msg.sender, amount);
    return bid_;
}
```

5.3 User Profile and Skill Management

Users connect their MetaMask wallet to access the platform. A user can create a profile and add relevant skills.

The profile system stores information such as skills, reviews, endorsements, and reputation. This allows users to present their capabilities and build a verifiable work history.

Code Snippet 5.3: Work Delivery — deliverJob()

```
function deliverJob(uint256 jobId, string calldata ipfsProof,
    string calldata description) external {
    Job storage job = jobs[jobId];
    require(msg.sender == job.selectedFreelancer &&
        job.status == JobStatus.InProgress, "Not allowed");
    require(job.milestoneCount == 0, "Use milestone submission");
    require(bytes(ipfsProof).length > 0, "Proof required");
    job.status = JobStatus.Delivered;
    job.deliveredAt = block.timestamp;
    job.deliveryProof = ipfsProof;
    job.deliveryDescription = description;
    job.revisionRequested = false;
    emit JobDelivered(jobId, msg.sender,
        block.timestamp + AUTO_RELEASE_PERIOD);
}
```

5.4. Escrow and Reward Management

The Escrow smart contract manages the reward associated with a job. Once a freelancer is selected, the agreed reward is secured through the smart contract. After the submitted work is approved by the client, the reward is released to the freelancer.

Code Snippet 5.4: Completion and Reward Release — completeJob()

```
function completeJob(uint256 jobId) external nonReentrant {
    Job storage job = jobs[jobId];
    require(job.client == msg.sender, "Only client");
    require(job.status == JobStatus.InProgress ||
        job.status == JobStatus.Delivered, "Not completable");
    require(!job.revisionRequested, "Revision response pending");
    _completeJob(jobId);
}

function _completeJob(uint256 jobId) internal {
    Job storage job = jobs[jobId];
    job.status = JobStatus.Completed;
    uint256 bidAmt = bids[job.acceptedBidId].amount;
    userProfiles[job.selectedFreelancer].jobsCompleted++;
    userProfiles[job.selectedFreelancer].totalEarned += bidAmt;
    if (job.milestoneCount == 0 && escrowContract != address(0))
        IEscrow(escrowContract).releasePayment(jobId);
    emit JobCompleted(jobId, job.selectedFreelancer, bidAmt);
}
```

5.5. Work Submission and IPFS Storage

Freelancers can upload their completed work through the SkillBounty application. The work file is stored using IPFS through Pinata, and the generated IPFS content identifier (CID) is associated with the submission for client review.

5.6. Wallet and Blockchain Integration

MetaMask is used to connect users with the SkillBounty platform and authorize blockchain transactions. The frontend communicates with the deployed Solidity smart contracts through the user's wallet, while confirmed transactions are recorded on the Mantle Sepolia test network.

5.7. Deployment

The smart contracts are deployed on the Mantle Sepolia public test network using Hardhat. The frontend is connected to the deployed contract addresses, allowing users to perform job, bidding, work submission, and reward-related operations through the blockchain.

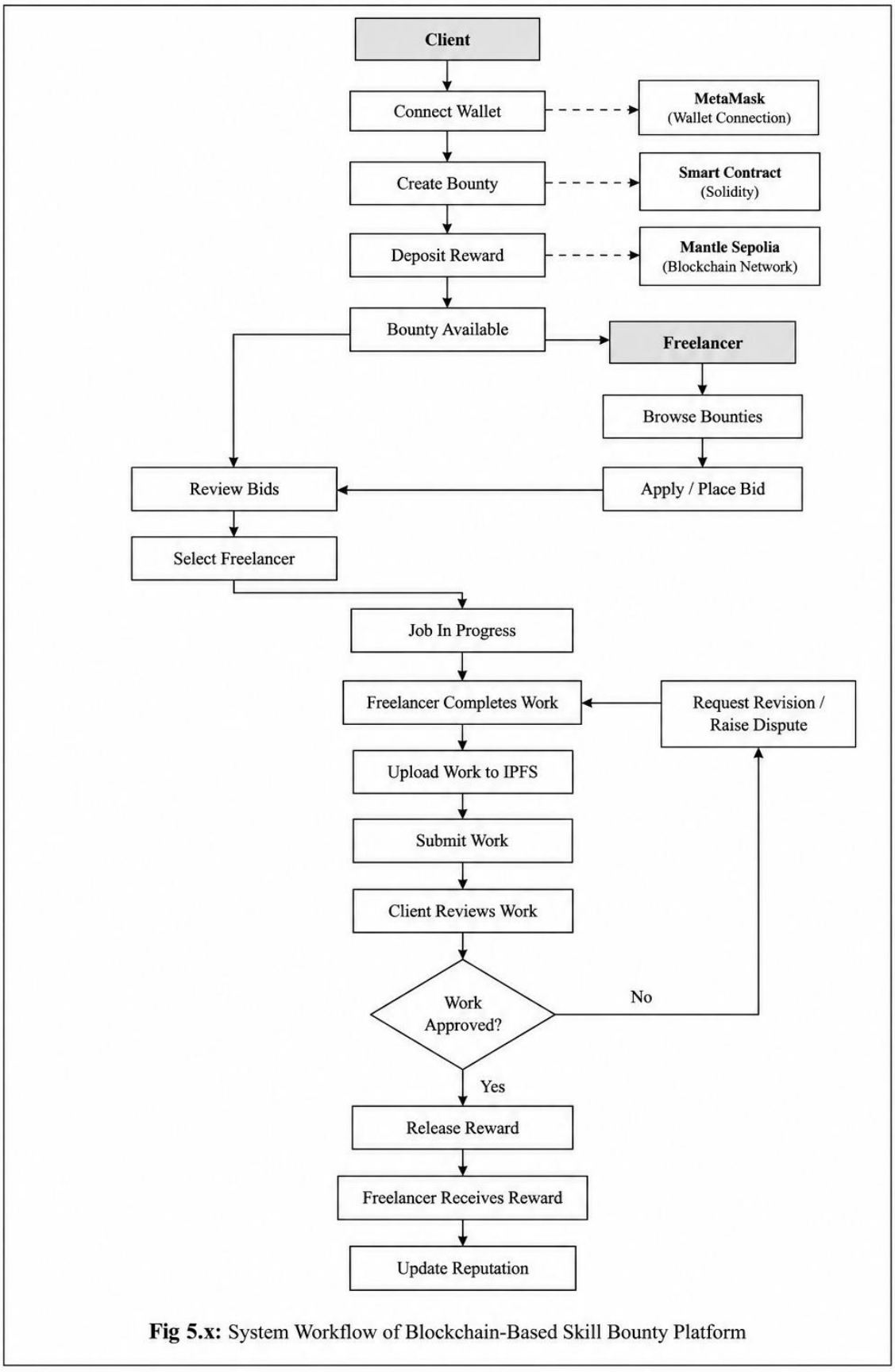


Fig 5.x: System Workflow of Blockchain-Based Skill Bounty Platform

6. BENEFITS

Blockchain technology enhances warranty management by introducing transparency, security, and automation. Traditional warranty systems often rely on centralized databases, which are prone to fraud, inefficiency, and data loss. By leveraging blockchain, warranty management becomes tamper-proof, cost-effective, and highly efficient. Below are the key advantages:

6.1 Transparency

Blockchain records important job, bidding, and payment transactions, allowing users to verify the activities performed through the platform.

6.2 Secure Reward Management

Smart contracts and escrow mechanisms help secure the agreed reward and release it according to the defined job workflow.

6.3 Reduced Intermediary Dependency

SkillBounty reduces dependency on a centralized authority for managing job transactions and reward settlement by using blockchain-based smart contracts.

6.4 Decentralized Work Storage

Completed work can be stored through IPFS using Pinata, providing a decentralized method for managing work deliverables.

6.5 Trust and Reputation

User skills, reviews, and reputation can be maintained through the blockchain-based profile system, helping clients evaluate freelancers based on their previous work.

6.6 Traceable Transactions

Blockchain transactions provide a verifiable record of important activities such as bounty creation, bidding, freelancer selection, and reward release.

6.7 Improved Freelancing Workflow

The platform provides a structured workflow from bounty creation and freelancer selection to work submission, approval, and reward distribution, making the overall process more transparent and organized.

7. CHALLENGES

While blockchain-based warranty systems offer numerous advantages, they also come with challenges and limitations that must be considered for effective implementation. Below are some of the key challenges:

7.1. Scalability Issues

As the number of users and blockchain transactions increases, the platform may face scalability challenges. Frequent blockchain interactions can also increase transaction processing time and network usage.

7.2. Smart Contract Security

Smart contracts control important operations such as job management and reward release. Any programming error or vulnerability in the contract logic may affect the correctness of these operations.

7.3. Wallet and Key Security

SkillBounty depends on MetaMask for user authentication and transaction authorization. Users must securely manage their wallet and private keys to prevent unauthorized access.

7.4. Privacy

Blockchain transactions are transparent and can be publicly verified. Therefore, sensitive user or job information should not be unnecessarily stored directly on the blockchain.

7.5. IPFS Availability

Work deliverables are stored using IPFS through Pinata. Reliable access to uploaded files depends on proper pinning and availability of the IPFS content.

7.7. Work Verification

Blockchain can record and enforce the payment process, but it cannot independently determine whether the submitted work is correct or meets the client's requirements. The client or authorized reviewer must verify the work before approving the reward.

8. CONCLUSION

The **Blockchain-Based Skill Bounty Platform** provides a decentralized approach to connecting clients with skilled freelancers. By using Solidity smart contracts, the platform supports transparent job management, bidding, escrow-based reward handling, work submission, and payment release.

The integration of **MetaMask, Mantle Sepolia, and IPFS through Pinata** provides secure wallet interaction, verifiable blockchain transactions, and decentralized work-file storage. The platform reduces dependency on centralized intermediaries while improving transparency and trust between clients and freelancers.

8.1. Future Outlook for Enhancements

In the future, SkillBounty can be enhanced with advanced skill-matching mechanisms to connect clients with suitable freelancers more efficiently. Improved profile analysis and recommendation features can help users identify opportunities based on their skills, experience, and previous work.

The platform can also be extended with automated work verification and improved dispute resolution mechanisms. External testing or verification services can be integrated to provide additional evidence before a reward is released.

Further development can include support for multiple blockchain networks and improved scalability. This can help the platform handle a larger number of users and transactions while maintaining efficient job and reward management.

A mobile application and enhanced privacy features can also be introduced to improve accessibility and user experience. These enhancements can make SkillBounty more reliable, scalable, and suitable for wider adoption in decentralized skill-based marketplaces.

9. SDG's ADDRESSED

The Blockchain-Based Skill Bounty Platform aligns with several United Nations Sustainable Development Goals (SDGs) by promoting decent work opportunities, encouraging technological innovation, improving transparency, and supporting equal access to digital employment. The platform uses blockchain technology and smart contracts to create a secure and transparent environment for connecting clients and skilled freelancers. Below are the key SDGs addressed and their justifications:

SDG 8 – Decent Work and Economic Growth

SkillBounty supports decent work and economic opportunities by connecting clients with skilled freelancers through a transparent digital platform. It provides individuals with opportunities to showcase their skills, complete jobs, and earn rewards.

SDG 9 – Industry, Innovation and Infrastructure

The project promotes technological innovation by applying blockchain, smart contracts, and decentralized storage to the freelancing domain. It demonstrates how emerging technologies can be used to develop secure and transparent digital infrastructure.

SDG 10 – Reduced Inequalities

SkillBounty can provide equal access to work opportunities based on skills rather than geographical location. Freelancers can participate in the platform and compete for opportunities using their profiles, skills, and reputation.

SDG 16 – Peace, Justice and Strong Institutions

Blockchain-based transaction records improve transparency and accountability in job and reward management. Smart contracts provide predefined rules for transactions, helping reduce uncertainty and build trust between clients and freelancers.

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11. APPENDIX A

https://drive.google.com/drive/folders/1G-JTSAmdWtfatpIjNsW_JjIY0jca5_tO?usp=sharing